
CONSERVATIVE CFD MODELLING OF IODINE DEPOSITION IN THERMAL-GRADIENT COLUMNS: NUMERICAL VERIFICATION AND INPUT SENSITIVITY

WORKING MANUSCRIPT FOR AUTHOR REVIEW

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ABSTRACT

Interpreting iodine deposition in thermal-gradient columns requires a consistent treatment of transport, gas speciation and exchange with condensed wall inventories. We present a finite-volume model in OpenFOAM for nine Pb–Bi–I gases in a frozen helium carrier, with element-conserving gas equilibrium and shared wall reservoirs. Reversible iodide exchange uses a Hertz–Knudsen–Schrage law coupled to the diffusive resistance between a cell centre and its wall face. Published thermosublimatography measurements supply three temperature profiles and deposition scans; unreported source histories and flow-reference conditions are treated as assumptions. Fifteen baseline and 64 follow-up simulations separate numerical refinement from selected input sensitivities. For the 60 s follow-up runs, the largest raw element-balance residue is 8.85×10^{-13} of supplied material. Halving the time step from 2 to 1 ms changes iodine deposition profiles by at most 0.323%, whereas refining from 18,432 to 41,472 cells changes them by 2.06–10.58%; spatial convergence remains unresolved. A $+29 \text{ kJ mol}^{-1}$ BiI gas Gibbs-energy perturbation changes the refined steel-case BiI₃ share of deposited iodide iodine from 94.57% to 2.00%, demonstrating strong conditional thermochemical sensitivity. Two shifts derived from discrepancies between the published silica temperature curve and its peak labels reduce 4–5 cm position errors to 0–1 cm, but do not establish a corrected experimental coordinate system. The results support a conservative computational method and an explicit assessment of uncertain inputs; they do not establish independent predictive validation or reproduce the three-hour experiment and subsequent cooling.

Keywords Iodine deposition · Thermosublimatography · OpenFOAM · Gas equilibrium · Element conservation · Input sensitivity

1 Introduction

Iodine released from lead–bismuth eutectic (LBE) can be transported in different chemical forms before depositing on cooler surfaces. The chemical form matters because it determines both the gas-phase inventory available for transport and the equilibrium pressure governing wall exchange. Liu et al. [1] investigated this behaviour by evaporating pure iodides and iodine-containing LBE into helium, separating the products along a negative temperature gradient and characterising the deposits. Their observations provide spatially resolved evidence for PbI₂ and BiI₃ deposition, together with potassium-containing deposits in some LBE experiments. Connecting such measurements to a transport calculation requires more information than a single deposition temperature: the temperature-coordinate mapping, carrier flow, source history and gas composition all enter the predicted distribution.

Lobresco et al. [2] compared temperature-based and vapour-pressure-driven CFD deposition models. The former require calibration of a deposition law; the latter connect deposition to vapour-pressure information through interfacial kinetics. Their study also motivates extending transport models to multiple chemical species and thermodynamic equilibrium. A multispecies calculation introduces an additional consistency requirement. Changes in chemical form

Table 1: Published configurations used for the baseline matrix. Masses refer to the initial sample, not to the amount released in a 60 s run. All experimental evaporation periods were 10,800 s.

Identifier	Column	Diameter (mm)	Sample (mg)	He (mL/min)
PbI2_SS	Steel	4.8	14.6	100
LBE-I_SS_II	Steel	4.8	346.0	45
LBE-I_SiO2_I	Silica	5.0	368.3	45

must preserve the transported elements, and different gas pathways leading to the same condensed material must use one physical wall inventory. An internally consistent implementation can nevertheless give uncertain experimental predictions if the thermodynamic or source inputs are poorly constrained.

The present work addresses these issues in the finite-volume framework of OpenFOAM [3]. Nine Pb–Bi–I gases are transported in a prescribed helium carrier. Homogeneous gas equilibrium redistributes the local elemental inventory, while separate interfacial channels exchange material with immobile condensed wall reservoirs. The wall flux includes both interfacial kinetic conductance and diffusion through the distance from the neighbouring cell centre to the wall. This avoids representing deposition by an arbitrarily chosen volumetric wall-layer thickness. It does not remove spatial discretisation error.

Equilibrium-based reactive transport is also supported by general thermodynamic packages such as GEM-Selektor and its GEMS3K kernel [4]. The software developed here contains a GEMS3K interface, but the calculations reported in this paper use a specialised ideal-gas element-potential solver with tabulated formation constants. We therefore distinguish the thermodynamic formulation from the particular numerical engine used in the reported simulations. Condensed phases are handled through wall exchange rather than through homogeneous cell equilibrium.

The study uses the published paper as its experimental input. No author-supplied temperature files, release histories or mass-flow-controller reference conditions are available. Instead of inferring a unique physical source from deposition data, we report the consequences of two explicit source constructions and selected variations in flow, diffusion, interfacial accommodation and thermodynamic information. Separate time-step and mesh studies identify which conclusions survive numerical refinement. A temperature-coordinate discrepancy in the silica experiment is examined as an input-consistency problem, with alternate mappings defined from published quantities before comparing the resulting CFD profiles.

The contribution is a documented conservative transport–equilibrium–wall coupling and a completed numerical and input-sensitivity assessment. The evidence comprises 15 baseline and 64 follow-up simulations of three available experimental configurations. These are computational cases, not independent experimental replicates. The scope is deliberately restricted to the executed model: a fixed carrier and temperature field, prescribed release, ideal-gas specification, and wall deposits without aerosol formation or surface-material-specific chemistry. Experimental comparisons are conditional assessments of that model, rather than a claim of independent predictive validation.

2 Experimental basis and input reconstruction

2.1 Available configurations

The experimental columns of Liu et al. [1] were 1.25 m long, with an inner diameter of 4.8 mm for 316L stainless steel and 5.0 mm for fused silica. A 15 mm sample boat was inserted into a hot starting furnace, followed by a furnace providing a decreasing temperature profile. Helium carried the evaporated material downstream. The source was withdrawn after three hours; the furnaces and gas flow were then turned off and the column cooled before deposit analysis. Temperature measurements and radiotracer scans used 1 cm increments.

Table 1 lists the three configurations for which both temperature curves and iodine scans could be reconstructed. The stand-alone BiI₃ experiment at 800 °C was considered in case preparation but was not run, because its required axial temperature profile was not available in the supplied figures. The missing case is not assigned a surrogate profile or counted as an unsuccessful simulation.

The two LBE samples had reported iodine mole fraction 9.04×10^{-4} and a 900 °C starting furnace. Their reported deposited iodine amounts were 0.191 and 0.203 mg, respectively. These deposited amounts constrain the prescribed mean iodine source in the present cases; they are not direct measurements of a time-resolved evaporation rate or an independent inlet-speciation measurement.

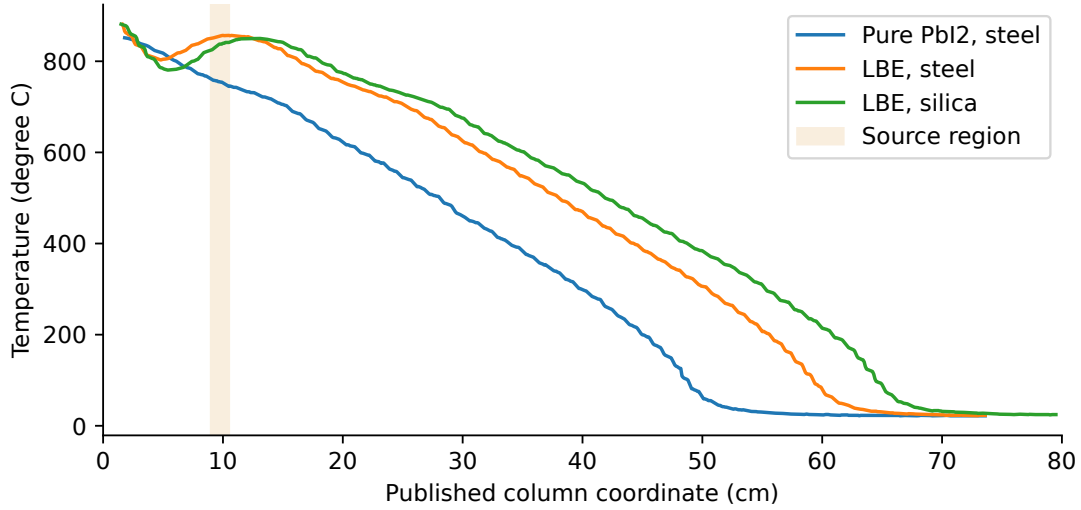


Figure 1: Temperature curves digitised from Liu et al. [1], with the assumed computational source interval. Unplotted parts of the CFD domain use constant endpoint extensions. The curves are experimental figure reconstructions, not temperature fields predicted by a conjugate heat-transfer calculation.

2.2 Digitisation and coordinate assumptions

Axial temperature curves and iodine histograms were digitised from the published figures. Temperatures are linearly interpolated at cell centres and wall faces; values beyond the plotted interval are held at the nearest endpoint. The computational coordinate is identified with the plotted coordinate. Its origin relative to the physical boat and furnace has not been independently confirmed. Figure 1 displays the reconstructed curves. These assumptions are retained in the original baseline matrix and varied explicitly in the silica-coordinate study.

The recorded digitisation resolutions are approximately 1 mm in position, 4 K in temperature and 0.4 percentage points in histogram height. They describe raster extraction, not the full experimental uncertainty. They do not include coordinate registration, detector response, occluded curve segments, temperature reproducibility or calibration. Likewise, the uncertainties printed alongside experimental deposition temperatures must not be interpreted as uncertainties on every digitised profile bin. The reconstructed histogram sums are 99.17%, 96.70% and 102.24% for pure PbI_2 , steel LBE and silica LBE. The primary comparison keeps those values without silently renormalising them.

In the LBE scans, the paper assigns 6.6% and 7.0% of iodine to KI, which is outside the Pb–Bi–I species set. The model profile is scaled to the remaining 93.4% or 93.0%, while the measured histogram retains all its published deposits. A second, explicitly conditional comparison normalises both profiles over their common scanned range (section 6). Neither comparison supplies the omitted potassium chemistry.

2.3 Simulation duration and measurement protocol

All source-driven cases run for 60 s while retaining a source rate based on the 10,800 s experimental period. Thus they release 1/180 of the assumed three-hour inventory; the experimental inventory is not compressed into a one-minute calculation. Continuous-source stationarity can inform deposition rates and conditional shapes, but cumulative wall inventories retain the initial filling transient. Four additional cases stop the source and carrier flow at 60 s and continue to 80 s under the original temperature profile. This probes residual-gas relaxation, not the experimental cooling history.

3 Transport, speciation and wall exchange

3.1 Frozen carrier and trace-species equations

The carrier fields ρ , $\mathbf{u}$, T and p are prepared from a steady laminar helium calculation and then held fixed. The reconstructed axial temperature is imposed throughout each cross-section. A gas species i is represented by Y_i , its

mass per unit carrier mass, and satisfies

$$\frac{\partial(\rho Y_i)}{\partial t} + \nabla \cdot (\rho \mathbf{u} Y_i) = \nabla \cdot (\rho D_i \nabla Y_i) + S_i^{\text{rel}} + S_i^{\text{wall}}. \quad (1)$$

Gas equilibrium is applied as a local redistribution between transport steps. Species do not alter the carrier density, momentum or temperature. The represented gases are Pb, PbI, PbI₂, Bi, Bi₂, BiI, BiI₃, I and I₂. Helium is inert. Condensed PbI₂, BiI₃, Pb and Bi are stored as immobile wall inventories. The solid-state identifiers in the code label these reservoirs; the equilibrium-pressure tables select between crystalline and liquid states where applicable.

No bulk nucleation, aerosols, thermal diffusion, hydrogen/oxygen chemistry, latent-heat feedback or chemical reaction with the steel or silica substrate is represented. Consequently, differences between the column cases arise from their prescribed thermal profiles, dimensions and source inputs, rather than a fitted substrate-specific sticking law.

3.2 Homogeneous ideal-gas equilibrium

Let ν_{ki} be the number of atoms of element $k \in \{\text{Pb}, \text{Bi}, \text{I}\}$ in gas i . At fixed local temperature the transported element concentrations are

$$b_k = \sum_i \nu_{ki} \frac{\rho Y_i}{M_i}. \quad (2)$$

The equilibrium solver seeks gas partial pressures and dimensionless element potentials λ_k satisfying

$$\frac{p_i}{p^\circ} = K_{f,i}(T) \exp\left(\sum_k \nu_{ki} \lambda_k\right), \quad \sum_i \nu_{ki} \frac{p_i}{RT} = b_k, \quad p^\circ = 10^5 \text{ Pa}. \quad (3)$$

The tabulated $K_{f,i}$ use a common elemental reference and the stated pressure standard. Concentrations convert back through $Y_i = M_i p_i / (\rho RT)$. This enforces elemental rather than individual species conservation during re-speciation. The helium pressure is not an additional reactive element constraint; its field remains prescribed under the trace-species approximation.

The executed solver uses logarithmic variables and a safeguarded Newton iteration. Pb and Bi subproblems are reduced at fixed iodine potential, with a Schur-complement treatment of the remaining coupling. Canonical initialisation avoids dependence on an unsaved equilibrium warm state when restarting a case. Re-speciation is requested every time step for cells with $T \geq 350 \text{ K}$; below this threshold the transported gas composition is retained. This cutoff is a numerical modelling choice, not a demonstration that reactions cease at 350 K. Homogeneous condensed phases are excluded, and instantaneous gas equilibrium substitutes for unmeasured finite reaction rates.

3.3 Reversible iodide exchange

For PbI₂ and BiI₃, the kinetic wall law uses a positive flux from condensate into gas:

$$j_i = G_i [p_{\text{eq},i}(T_w) - p_{i,w}], \quad G_i = C_e s_i \frac{2\alpha_i}{2 - \alpha_i} \sqrt{\frac{M_i}{2\pi RT_w}}. \quad (4)$$

Here j_i is in $\text{kg m}^{-2} \text{ s}^{-1}$, M_i is in kg mol^{-1} , pressures are in Pa, α_i is the accommodation coefficient and C_e , s_i are dimensionless multipliers. The baseline uses $C_e = s_i = \alpha_i = 1$. This is the Hertz–Knudsen–Schrage form used in vapour-pressure-driven deposition modelling [2]. The condensed-state pressure is calculated from the gas and condensed Gibbs energies, using the stable tabulated condensed branch. The wall temperature, rather than a prescribed deposition threshold, selects that pressure.

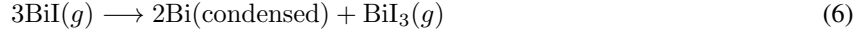
The kinetic law must also communicate with the cell-centre concentration through molecular diffusion. With d_f the cell-centre-to-wall distance, $p_{i,c} = \chi_{i,c} Y_{i,c}$ and $\chi_{i,c} = \rho_c RT_c / M_i$, elimination of the wall concentration gives

$$G_{i,f}^{\text{eff}} = \frac{G_i}{1 + G_i \chi_{i,c} d_f / (\rho D_i)_f}, \quad j_{i,f} = G_{i,f}^{\text{eff}} [p_{\text{eq},i}(T_w) - \chi_{i,c} Y_{i,c}]. \quad (5)$$

This combines kinetic and half-cell diffusion resistances in series. The gas equation receives the face flow $A_f j_{i,f}$ as an implicit cell contribution, while the wall reservoir receives its negative. Evaporation is limited by the available face inventory over the time step; a bare wall cannot evaporate. Inventory updates use the exchange actually realised by the final gas solve, including active bounds, rather than an independently evaluated explicit predictor. Reversal between deposition and evaporation therefore does not create material.

3.4 Shared metal inventories and a reactive channel

Additional wall channels connect Pb(g) to condensed Pb and Bi(g) and Bi₂(g) to a common condensed Bi inventory, with their own equilibrium pressures and stoichiometric mass conversion. They use the baseline interfacial coefficients. The reactive channel



is enabled only in the forward direction in the executed cases. For unit activity of condensed Bi, the dimensionless reaction constant and effective reactant equilibrium pressure are

$$K_r(T_w) = \exp \left[-\frac{2G_{\text{Bi},c}^\circ + G_{\text{BiI}_3,g}^\circ - 3G_{\text{BiI},g}^\circ}{RT_w} \right], \quad (7)$$

$$p_{\text{eq,BiI}} = p^\circ \left[\frac{p_{\text{BiI}_3,c}/p^\circ}{K_r(T_w)} \right]^{1/3}, \quad (8)$$

$$j_{\text{BiI}} = \min\{0, G_{\text{BiI}}^{\text{eff}}[p_{\text{eq,BiI}} - p_{\text{BiI},c}]\}. \quad (9)$$

All Gibbs energies in K_r are evaluated at T_w . This is a phenomenological HKS-conductance treatment of the reaction driving force, with the same half-cell resistance as the other channels. The coproduct pressure is evaluated from the beginning-of-step gas field, and the minimum suppresses reverse reaction. The formula consumes three BiI molecules, supplies one BiI₃ molecule to the gas and adds two Bi atoms to the shared wall reservoir. All channels access that same Bi inventory; a separate inventory for each reaction would duplicate a physical material. Coproducts are applied after the transported reactants have been solved. This irreversible channel is a model assumption and must not be interpreted as a fully reversible surface-equilibrium model.

3.5 Prescribed release constructions

Source model Q1 releases PbI₂ and, for LBE, BiI₃ at constant rates

$$q_{1,i} = \frac{m_i}{M_i \tau_{\text{exp}}}, \quad \tau_{\text{exp}} = 10,800 \text{ s}. \quad (10)$$

For pure PbI₂, $m_i = 14.6 \text{ mg}$ is the initial sample mass; complete uniform release over the experimental period is assumed. For steel LBE, $m_{\text{PbI}_2} = 0.111 \text{ mg}$ and $m_{\text{BiI}_3} = 0.181 \text{ mg}$; for silica LBE, they are 0.243 and 0.085 mg. These compound masses were inferred from deposited material in the experimental paper. Their use conditions both iodine loading and much of the Q1 species split on the measurements.

Source model Q2 supplies atomic Pb, Bi and I to the gas-equilibrium system. With q_{He} the full-column helium molar flow and p_h the source pressure, its elemental rates are

$$q_{2,\text{Pb}} = q_{1,\text{PbI}_2} + q_{\text{He}} \beta p_{\text{Pb}}(T_h)/p_h, \quad (11)$$

$$q_{2,\text{Bi}} = q_{1,\text{BiI}_3} + q_{\text{He}} \beta [p_{\text{Bi}}(T_h) + 2p_{\text{Bi}_2}(T_h)]/p_h, \quad (12)$$

$$q_{2,\text{I}} = 2q_{1,\text{PbI}_2} + 3q_{1,\text{BiI}_3}. \quad (13)$$

The metal pressures use ideal LBE activities $a_{\text{Pb}} = 0.447$ and $a_{\text{Bi}} = 0.553$ at $T_h = 1173.15 \text{ K}$. The multiplicative metal saturation β is unmeasured. Q2 changes the metal supply and allows the iodide split to emerge from the model, but still fixes its iodine source from deposited iodine. It is therefore a source hypothesis, not independently identified LBE evaporation. Rates are distributed over an assumed source region without resolving a liquid surface, depletion of the physical boat or its obstruction of the flow.

4 Numerical implementation and study design

4.1 Geometry and carrier preparation

A structured axisymmetric 5° wedge represents the 1.25 m column (fig. 2). The steel and silica radii are 2.4 and 2.5 mm, respectively. The baseline mesh has $384 \times 48 \times 1 = 18,432$ cells. Coarse and fine meshes have $256 \times 32 \times 1 = 8,192$ and $576 \times 72 \times 1 = 41,472$ cells, refining axial and radial spacing by a factor of 1.5 between successive levels. The source spans $x = 0.09\text{--}0.105 \text{ m}$ and the radial interval $r = 0.25R\text{--}0.75R$. Its position and selection are computational assumptions; the physical boat is omitted from the carrier geometry. Full-column source and inlet rates are multiplied by $5/360 = 1/72$ for the wedge.

The inlet flow rates are converted to helium mass flow assuming a controller reference of 298.15 K and 101,325 Pa. Full-column rates are $2.727 \times 10^{-7} \text{ kg s}^{-1}$ at 100 mL/min and $1.227 \times 10^{-7} \text{ kg s}^{-1}$ at 45 mL/min. The reference

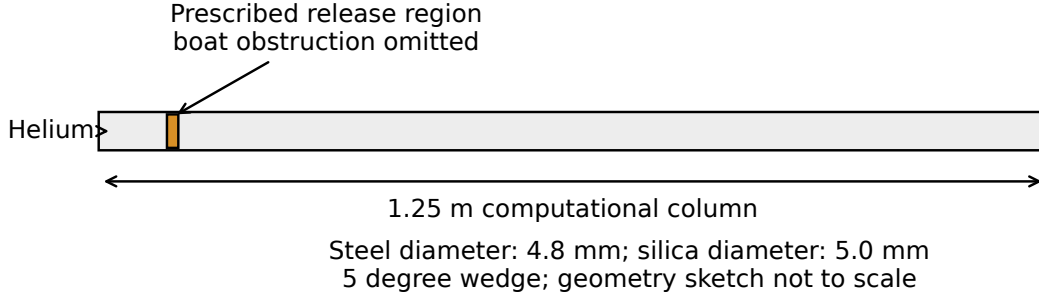


Figure 2: Computational geometry and prescribed source region. The source is distributed in selected cells; the sketch does not represent a resolved boat surface. Tube length and material-specific diameters are retained, and the 5° wedge supplies the axisymmetric reduction.

conditions are not specified by the supplied paper. A separate study uses 273.15 K at the same reference pressure. Helium uses the perfect-gas law, constant heat capacity and Sutherland viscosity (appendix A). Steady laminar carriers are prepared with rhoSimpleFoam; the reconstructed temperature profile remains prescribed.

The converged face mass flux is retained instead of reconstructing it from interpolated velocity. At these low mass flows, residual discrete divergence is removed by a zero-initialised potential correction before species transport. Carrier preparation rejects an original cell or global flux defect exceeding 10^{-6} of inlet flow, or a flux/velocity correction exceeding 10^{-5} of the corresponding scale. After correction, inlet, global and cell continuity and impermeable-boundary leakage are audited against 10^{-12} of inlet flow. This procedure makes the frozen transport field discretely conservative; it is not a new physical flow model or evidence of convergence of the velocity field.

4.2 Discretisation and coupling

The transient species equations use implicit Euler time integration, first-order upwind advection and corrected Gauss linear diffusion. All nine transported gases use the same advection scheme. The baseline time step is 2 ms. Wall exchange is assembled implicitly with nonnegative source and sink coefficients and shared inventory bounds. Configured species correctors and active-bound guards reconcile the final gas solution with the realised wall transfer. Gas re-speciation then redistributes species under the local element constraints. This sequence is first order in time; a conservative equilibrium projection alone does not eliminate splitting or transport error.

The baseline configuration requests three species correctors with a 10^{-16} tolerance and up to eight bound guards with a 10^{-13} guard tolerance. These are stopping settings, not guarantees that every outer wall-regime iteration attains the requested tolerance. The final transport defect, bound status and material ledger must be examined independently (section 5.1). Additional short corrector replays isolate late-time iteration sensitivity but do not establish full-run iteration independence.

The baseline diffusivity is $D_i = 10^{-4} \text{ m}^2 \text{ s}^{-1}$ for every gas. Three transport alternatives halve or double this value for all gases, or replace only the PbI_2 coefficient by a temperature- and pressure-dependent PbI_2 –He estimate. The latter’s collision parameters are estimates, not independently measured binary transport data. Accommodation variations change the two reversible iodide pairs to $\alpha = 0.1$ or 0.01 , retaining the baseline metal-channel coefficients.

4.3 Completed case matrix

The 15 baseline cases comprise Q1 for each available configuration and Q2 for the two LBE columns, at $\beta \in \{0.001, 0.01, 0.1\}$ with two thermodynamic branches. The second branch adds a constant 29 kJ/mol to BiI(g) ’s standard Gibbs energy; it is a sensitivity surrogate, not a second experimentally qualified dataset. All completed calculations use the same solver executable, identified by its SHA-256 in the saved reports.

Table 2 describes the 64 follow-up cases. Mesh and time-step checks cover pure PbI_2 and three steel Q2 cases: baseline thermodynamics at 0.1% and 1% metal saturation and shifted BiI thermodynamics at 1%. Physical transport, accommodation, source-position and flow-reference variations cover pure PbI_2 , those two baseline steel Q2 cases, and silica Q2 at 0.1%. Extra source-loading cases use both columns and both thermodynamic branches. Temperature-

Table 2: Completed follow-up matrix. The baseline cases are additional references. Numerical comparisons use one changed resolution at a time; flow-stop cases continue a 60 s baseline to 80 s.

Study	Cases	Variation
Mesh and time step	12	Three grids; 2 and 1 ms
Gas diffusion	12	$D/2$, $2D$, PbI ₂ –He correlation
Iodide accommodation	8	$\alpha = 0.1, 0.01$
Source position	8	$\Delta x_s = \pm 1.5$ cm
Q2 metal loading	12	$\beta = 0.002, 0.003162, 0.005623$
Flow reference	4	273.15 versus 298.15 K
Silica temperature coordinates	4	$\Delta x_T = -4.10, -4.65$ cm
Flow-stop relaxation	4	60–80 s; fixed temperature

coordinate variants use silica Q1 and silica Q2 at 0.1%. This is a structured set of conditional comparisons, not a factorial uncertainty study.

The source-position alternatives move the release region by ± 1.5 cm without adding a boat obstruction. Flow-stop cases restart the saved 60 s fields, remove injection, set carrier flow to zero and impose zero normal species flux at both ends. The original nonuniform temperature profile stays fixed. Their comparisons therefore measure redistribution of residual gas during a 20 s fixed-temperature flow stop, not a clean-helium purge or a full cooling calculation.

5 Numerical verification and resolution limits

5.1 Completion, material balance and solver diagnostics

All 15 runnable baseline cases and all 64 follow-up cases completed with their configured numerical acceptance checks. A PASS classification means that the computation completed and passed those checks; it is not an experimental-validation classification. Element conservation is examined both through the solver ledger and through an independently formed material residue. For element k the latter is

$$R_k = N_k^{\text{gas}} + N_k^{\text{wall}} + N_k^{\text{sample}} + N_k^{\text{net out}} + N_k^{\text{removed}} - N_k^{\text{initial}} - N_k^{\text{released}}, \quad (14)$$

where N_k are molar atom inventories and outward transport is signed. The ledger separately records the linear-solve defect and any clamping correction. These terms explain bookkeeping closure but are not inserted into R_k when assessing the raw material balance.

Across recorded steps and represented elements, the baseline’s largest raw residue divided by supplied material is 5.97×10^{-13} . For the 60 source-driven follow-up cases it is 8.85×10^{-13} ; the largest accounted closure is 9.60×10^{-16} and the largest recorded solver defect is 8.64×10^{-13} on the same supplied-material scale. The four flow-stop cases retain their restart ledger and are excluded from this newly accumulated-source summary. These results support conservation of the executed discrete model. They do not quantify physical accuracy or discretisation error.

The baseline logs report no gas-equilibrium solver failures and a maximum equilibrium element residue of 6.43×10^{-16} . Input-validity and solver-success counters are distinct: startup sign flags caused equilibrium to be skipped in up to 1,170 cells on one step. The maximum negative iodine diagnostic was 1.14×10^{-124} of supplied iodine. Matched five-step startup replays identified extremely small leading-tail inventories at these flags, with no flagged cells remaining at the final step. Thus the calculations should not be described as having identically zero negative trace values, even though the reported amount is immaterial to the macroscopic balance. The sum of individual species mole-fraction maxima was at most 9.19×10^{-5} ; since these maxima need not coincide, this is an upper bound on the reactive trace loading, not a measured mixture mole fraction at one location.

Some paired wall-coupling iterations did not meet the requested 10^{-16} outer criterion. This is reported separately from equilibrium failure, the final transport defect and inventory-bound checks. Matched restarts of four representative cases at 59 s compared three and eight correctors and outer tolerances of 10^{-16} and 10^{-14} for 20 time steps, or 40 ms. The largest species-field change, scaled by its control maximum, was below 3.6×10^{-8} , with negligible new wall-increment changes. This supports a residual-floor interpretation at the tested checkpoints, but leaves full-run iteration sensitivity unqualified. Planned manufactured-solution and cross-code comparisons are not presented as completed verification evidence.

Table 3: Iodine inventory-profile differences from the 18,432-cell, 2 ms reference. All entries are percentages. Coarse and fine meshes have 8,192 and 41,472 cells at 2 ms; ΔN_I is fine minus medium total deposited iodine, divided by the medium total.

Case	2 to 1 ms	Coarse/medium	Fine/medium	ΔN_I
Pure PbI ₂	0.0043	6.57	5.06	-0.0028
Steel Q2, 0.1%	0.2326	9.02	5.26	-0.0016
Steel Q2, 1%	0.3228	9.72	2.06	-0.0133
Steel Q2, 1%, shifted BiI	0.1536	10.47	10.58	-0.0064

5.2 Separate time-step and mesh comparisons

For profiles on the common 1 cm bins, let a_b be deposited iodine atoms in bin b for a comparison case and a_b^0 the corresponding baseline inventory. We report

$$E_{L1} = 100 \frac{\sum_b |a_b - a_b^0|}{\sum_b a_b^0}, \quad \Delta N_I = 100 \frac{\sum_b a_b - \sum_b a_b^0}{\sum_b a_b^0}. \quad (15)$$

These measures retain amount changes. Normalised shape curves are shown separately and must not be confused with this unnormalised profile difference. Source, physical coefficients, equilibrium engine and executable are fixed within each numerical comparison.

Table 3 shows that reducing the time step from 2 to 1 ms on 18,432 cells changes the iodine profile by 0.0043–0.3228%. The fine/medium mesh differences at a fixed 2 ms are 2.06–10.58%. The shifted-BiI steel case has essentially the same magnitude of profile change on successive refinements. Its spatial error is therefore not resolved by the time-step result. Total deposited iodine changes by less than 0.014% from medium to fine, and the centroid changes by at most 0.0652 cm. Stable integrals coexist with appreciable differences in local profile shape.

Three-grid scalar checks follow the distinction between refinement trends and an established asymptotic range [5]. For the 34 nontrivial or zero-valued combinations of species, total, centroid and width, 17 sequences are oscillatory, four have nondecreasing differences, one is unresolved at roundoff and 12 admit only conditional Richardson/GCI estimates. All four iodine-total sequences are oscillatory. The conditional estimates are retained in the accompanying data with their status; they are not promoted to confirmed uncertainty bounds. The current evidence does not establish mesh independence or a single observed convergence order for this coupled model.

5.3 Binned maxima and numerical diffusion

A deposition maximum selected on a 1 cm grid is particularly fragile when adjacent bins have similar heights. In pure PbI₂ the medium-grid maximum is at 37 cm and 350.7 °C, while the 36 cm bin is only about 0.05% lower. The fine-grid maximum moves to 36 cm and 366.5 °C despite a centroid shift of only −0.0585 cm. In the steel Q2 0.1% case, the dominant BiI₃ bin changes from 57 to 58 cm, moving its reported temperature from about 166.4 to 138.8 °C. A one-bin switch can therefore create a temperature change of nearly 28 K without a comparable displacement of the profile as a whole. Peak temperature alone is an insufficient convergence metric.

First-order upwind advection also adds spatial smoothing. The local one-dimensional estimate $D_{\text{num}} \simeq |u_x| \Delta x / 2$ is comparable to or greater than the prescribed $10^{-4} \text{ m}^2 \text{ s}^{-1}$ diffusivity in parts of the baseline carriers. This is a Taylor-series screening estimate, not a measured truncation error for the radial, variable-density coupled problem. It explains why halving the time step cannot establish convergence of profile widths. Further spatial refinement or a separately verified bounded higher-order transport scheme is needed before precise profile-shape prediction is claimed.

6 Conditional comparison with deposition measurements

6.1 Comparison measures

The primary comparison uses reconstructed relative iodine yields per 1 cm bin. Let e_b be the experimental percentage and let

$$m_b = (100 - y_{\text{KI}}) \frac{a_b}{\sum_{\text{full column}} a_b} \quad (16)$$

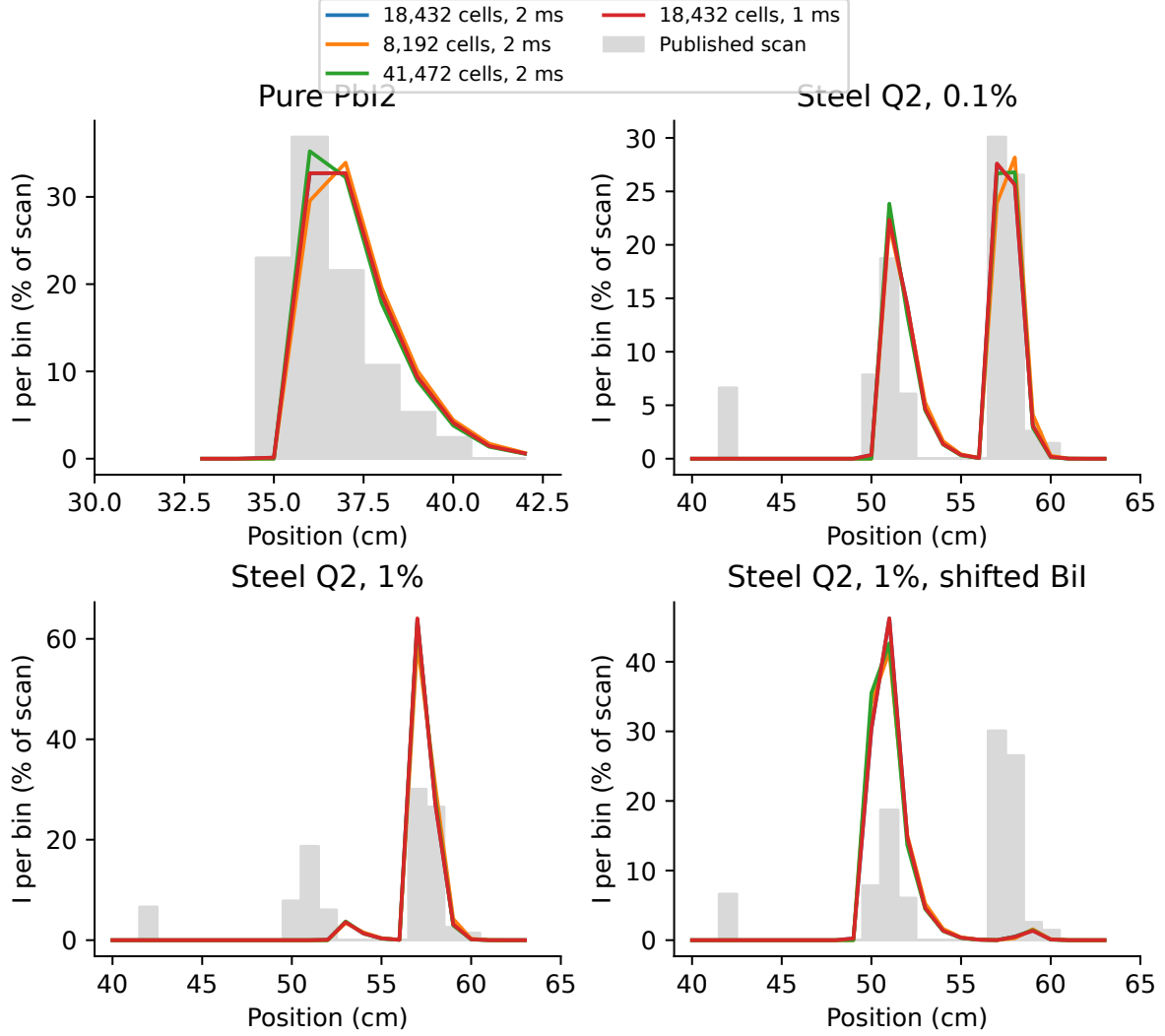


Figure 3: Separate mesh and time-step comparisons. Each curve is normalised over the common scanned interval for visual comparison; the unnormalised differences are in table 3. Grey histograms are reconstructed experimental iodine scans [1] and include any KI contribution.

be the modelled percentage, with the stated excluded KI yield $y_{KI} = 0, 6.6$ or 7.0 . Over the common scanned bins $\mathcal{B}$, we report the raw discrepancy

$$L_{\text{raw}} = \sum_{b \in \mathcal{B}} |m_b - e_b| \quad (\text{percentage points}). \quad (17)$$

This preserves the experimental digitisation sum and the model's explicit KI exclusion. It is neither an absolute deposited-mass error nor a percentage relative to an experimental uncertainty.

A secondary shape diagnostic sets $\hat{m}_b = m_b / \sum_{\mathcal{B}} m_b$ and $\hat{e}_b = e_b / \sum_{\mathcal{B}} e_b$, and calculates

$$O = 100 \sum_{b \in \mathcal{B}} \min(\hat{m}_b, \hat{e}_b). \quad (18)$$

Overlap O is conditional on the common scanned range and discards capture and total-yield differences. The experimental normalisation still contains KI. It is a descriptive shape measure, not a validation probability or a fitted goodness-of-fit statistic.

Table 4: Original 18,432-cell Q1 comparisons. Positions are in cm and temperatures in °C. Experimental temperature ranges are those printed by Liu et al. [1]; they are not statistical confidence intervals for model error.

Case	Species	x_{exp}	x_{CFD}	T_{exp}	T_{CFD}
Pure PbI ₂ , steel	PbI ₂	36	37	368±18	350.7
LBE, steel	PbI ₂	51	51	288±18	288.1
LBE, steel	BiI ₃	57	57	154±14	166.4
LBE, silica	PbI ₂	51	56	306±24	287.6
LBE, silica	BiI ₃	59	63	127±39	152.8

6.2 Q1 peaks and complete profiles

Table 4 compares the five available Q1 iodide maxima with the printed peak temperatures. All calculated peak temperatures lie within the stated temperature ranges. Pure PbI₂ is near the low-temperature edge of its range, and its binned maximum changes under refinement as described above. Steel LBE places both iodide maxima at the measured positions, while silica LBE places PbI₂ and BiI₃ respectively 5 and 4 cm downstream. Agreement with a printed temperature is therefore not sufficient to establish the correct spatial location.

Figure 4 retains the full profile comparison. The raw L_{raw} values are 53.90, 53.45 and 181.84 percentage points for pure PbI₂, steel LBE and silica LBE. Corresponding normalised overlaps are 72.93%, 71.88% and 7.20%. The silica case therefore has a large profile discrepancy even though both printed temperature ranges contain its calculated maxima.

The Q1 steel BiI₃ share of deposited iodide iodine is 65.34%, close to the measured iodide-only share of $61.3/(61.3 + 32.1) = 65.63\%$. For silica the corresponding values are 28.68% and $27/(27 + 66) = 29.03\%$. These agreements are substantially conditioned by the source masses inferred from those same deposits. They cannot serve as independent validation of gas speciation or LBE release. Likewise, source-driven calculations capture 91.27–98.20% of supplied iodine on the wall by 60 s, but that does not establish experimental three-hour capture: the remaining inventory, release history and post-release cooling are different questions.

7 Input sensitivity and interpretation

7.1 Source loading and thermodynamic information

The Q2 cases expose a sensitivity hidden by prescribing the iodide split. Figure 5 shows the BiI₃ iodine fraction

$$f_{\text{BiI}_3} = \frac{3N_{\text{BiI}_3}^{\text{wall}}}{2N_{\text{PbI}_2}^{\text{wall}} + 3N_{\text{BiI}_3}^{\text{wall}}} \quad (19)$$

This includes only iodine in the two modelled deposited iodides. It is distinct from the BiI₃ fraction of all experimental iodine, which also contains KI. Additional metal-saturation cases fill selected points between the original 0.1% and 1% branches; line segments connect computed points and are not a fitted release law.

At 1% metal saturation on the fine steel mesh, the baseline thermodynamics gives $f_{\text{BiI}_3} = 94.57\%$. Raising the BiI(g) standard Gibbs energy by 29 kJ/mol gives 2.00% with the same numerical resolution. The large difference survives refinement, demonstrating conditional thermodynamic sensitivity rather than numerical noise. The perturbation is applied consistently to formation and reactive-wall constants. A constant Gibbs-energy shift scales the formation constant by $\exp[-29,000/(RT)]$, approximately 0.051 at 1173 K and 0.00093 at 500 K. It is a substantial surrogate perturbation, not an independently measured correction to BiI thermodynamics.

Source loading also matters. With baseline thermodynamics the steel BiI₃ share at 0.2%, 0.316% and 0.562% metal saturation is 52.50%, 70.58% and 94.56%; the shifted branch gives 47.66%, 37.43% and 15.88%. The low-loading trend is not universally monotonic. The original steel 0.1% baseline branch gives 56.47% and an 81.10% scan overlap, making it a useful diagnostic comparison, but not an identified physical saturation. The published deposits alone cannot uniquely distinguish source loading, thermodynamic information and omitted chemistry. Choosing a branch because it best matches one scan would introduce calibration and require independent validation.

7.2 Silica temperature-coordinate discrepancy

The published silica scan places PbI₂ and BiI₃ maxima at 51 and 59 cm, labelled 306±24 and 127±39 °C. At those coordinates, the digitised temperature curve gives approximately 369 and 235 °C: differences of 63 and 108 K. On that

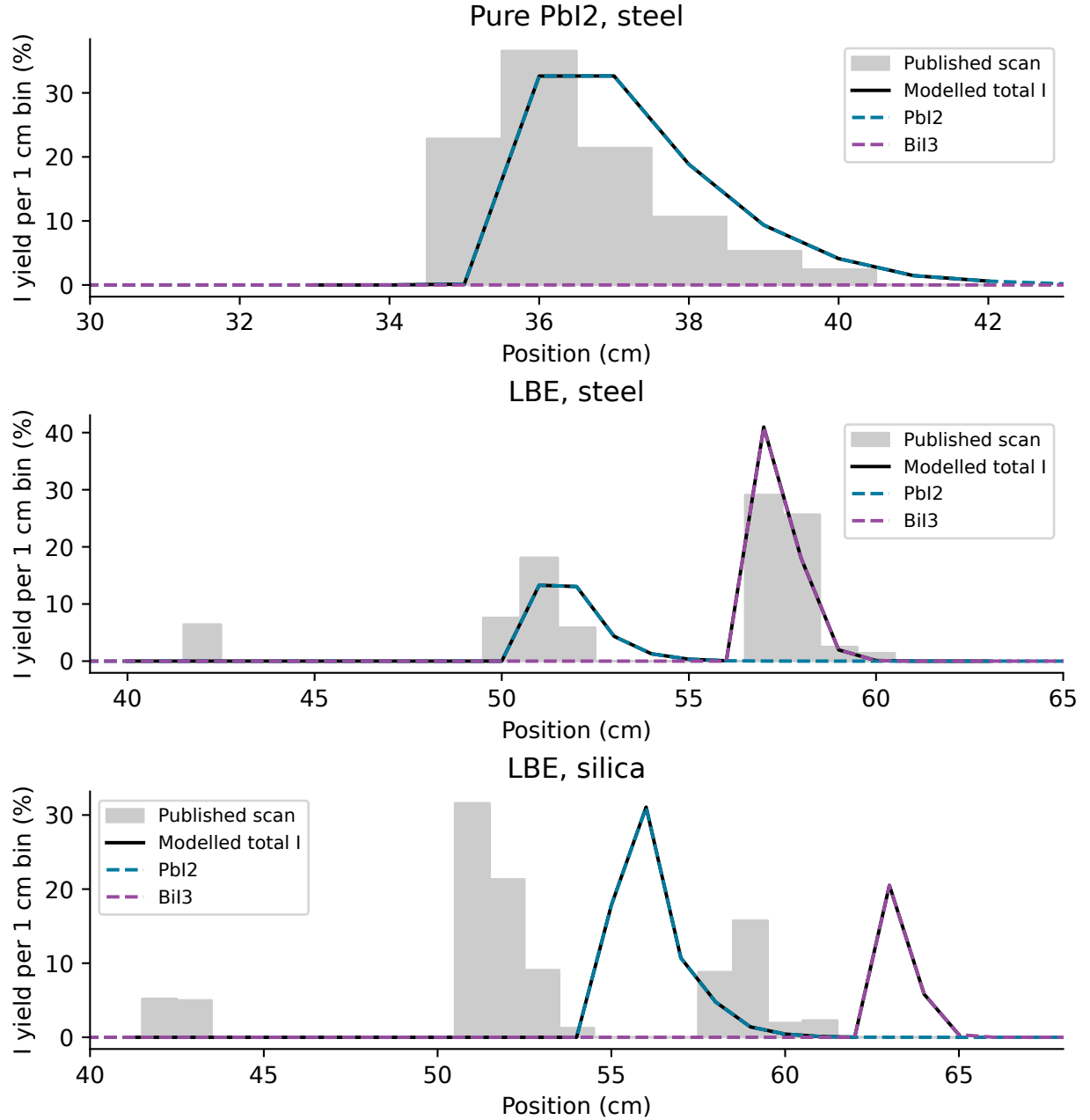


Figure 4: Original Q1 iodine profiles. Solid black lines are total modelled wall iodine; dashed lines show its two iodide contributions. Grey histograms are digitised published scans [1]. Model totals use 100%, 93.4% and 93.0% normalisations for pure, steel LBE and silica LBE, respectively, to expose the omitted KI yield; experimental bars retain their digitised values.

same curve, 306 and 127 °C occur at 55.096 and 63.653 cm. The discrepancy exceeds the nominal raster temperature resolution and suggests a temperature-coordinate consistency problem in the supplied inputs. It does not identify which measurement, annotation or registration requires correction.

Two diagnostic mappings shift only the silica temperature curve upstream by 4.096 or 4.653 cm, aligning one of those curve crossings with the corresponding labelled position. Equivalently, $T_\delta(x) = T_0(x + \delta)$ for $\delta > 0$, with $\Delta x_T = -\delta$. The gamma scan remains fixed. These shifts are derived from the published curve and labels, not optimised against the CFD profiles. All original unshifted results are retained.

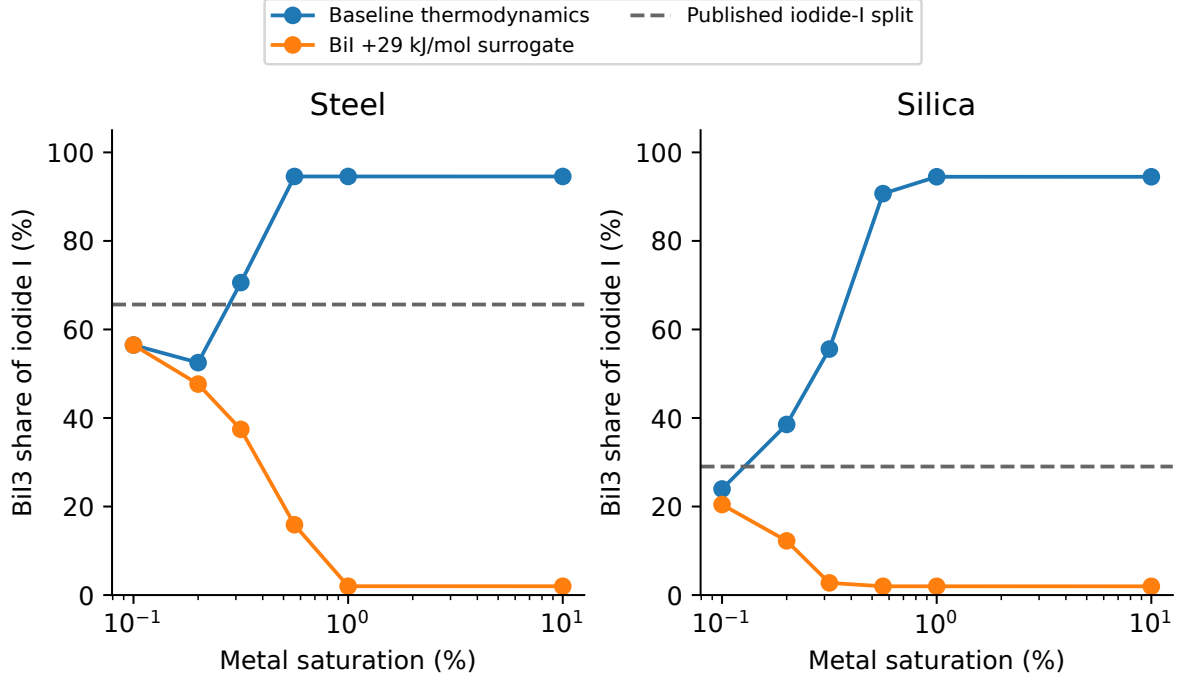


Figure 5: Conditional source-loading and BiI thermodynamic sensitivity for steel and silica LBE. Symbols denote completed Q2 cases; segments only connect those points. Dashed grey lines show the published BiI_3 share after excluding KI [1]. The +29 kJ/mol branch is a surrogate perturbation, not a second qualified thermodynamic dataset.

Table 5: Silica temperature-coordinate hypotheses. Shifts and peak positions are in cm; raw L1 is in percentage points and overlap in percent. Q2 uses baseline thermodynamics and 0.1% metal saturation. Every row uses the same fixed experimental scan.

Source	Δx_T	x_{PbI_2}	x_{BiI_3}	Raw L1	Overlap
Q1	0.00	56	63	181.84	7.20
Q1	-4.10	52	59	46.28	75.29
Q1	-4.65	51	58	37.05	81.00
Q2	0.00	56	63	181.84	7.20
Q2	-4.10	51	59	39.40	79.16
Q2	-4.65	51	59	34.31	83.33

Table 5 and fig. 6 shows that the shifts move Q1 maxima from 56/63 cm to 52/59 or 51/58 cm and improve normalised overlap from 7.20% to 75.29% or 81.00%. The low-loading Q2 variants place both maxima at 51/59 cm, with overlaps 79.16% and 83.33%. This explains why a plausible coordinate change has a larger effect than the tested source-position or time-step changes. It does not confirm either shift as the experimental truth. The two shifts also produce different peak temperatures and cannot simultaneously reconcile both original temperature labels by a uniquely established mapping.

The shifted low-loading Q2 cases retain a BiI_3 share near 23.97%, about five percentage points below the published iodide-only split. Correct peak positions therefore do not ensure the correct partition. Resolving the discrepancy requires the original thermocouple profile and its coordinate registration, rather than selecting the CFD variant with the highest overlap.

7.3 Transport, kinetics, flow and shutdown

Table 6 reports the range of iodine profile changes from the designated baseline within each tested category, using eq. (15). They are responses to finite hypothetical changes, not statistical confidence limits. The source-loading row uses its 1% reference branch; the temperature-coordinate row measures shifts from the original silica input and can

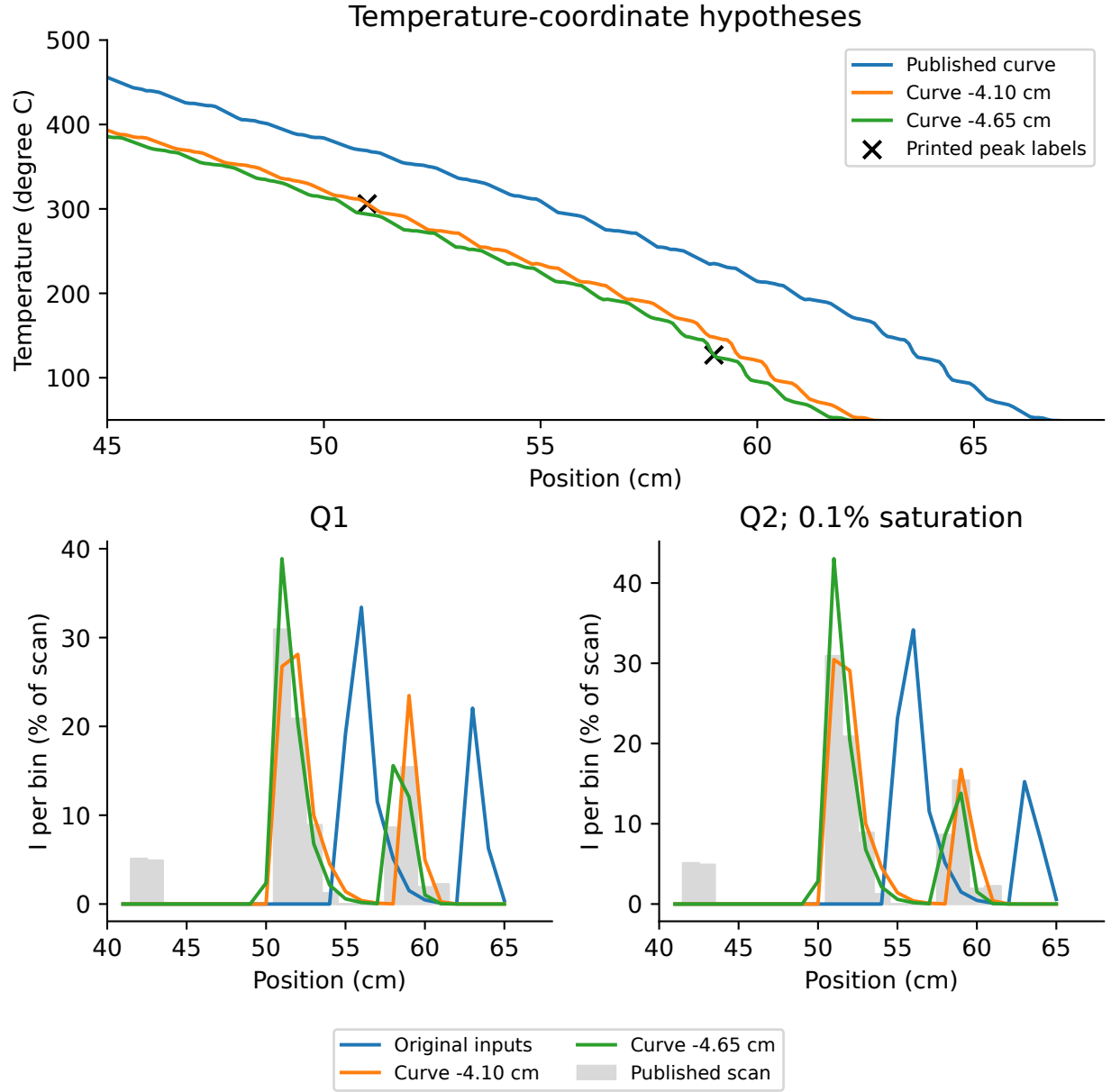


Figure 6: Input-derived silica mappings and their consequences. Top: the original reconstructed temperature curve, two shifts and printed peak labels. Bottom: common-scan-normalised iodine profiles for Q1 and low-loading Q2. Improved agreement is conditional on an unconfirmed coordinate hypothesis. Published data are adapted from Liu et al. [1].

approach 200% when profile supports move apart. These baseline-relative differences must not be confused with raw discrepancies against the experimental scan.

Diffusion alternatives change profiles by up to 14.20%, and a 273.15 K flow reference raises inferred helium mass flow by 9.15% relative to 298.15 K and changes profiles by up to 10.93%. Iodide accommodation changes give up to 4.21%, whereas moving the prescribed source region by ± 1.5 cm gives at most 0.19%. This small source-position response concerns the selected empty-tube source model; it cannot rule out an effect of a real boat on flow or thermal fields. The maxima for these categories are shown in fig. 7 and should be compared with the unresolved 2.06–10.58% fine/medium spatial differences. Where numerical and input changes have comparable magnitude, the present results do not rank their physical importance precisely.

Table 6: Ranges of baseline-relative iodine inventory-profile changes (percent) over each completed sensitivity category. Shutdown compares 80 s inventories with their 60 s restart state; other rows compare 60 s inventories. Case counts and variations are in table 2.

Assumption	Minimum	Maximum
Gas diffusion	0.15	14.20
Iodide accommodation	0.25	4.21
Source position	0.08	0.19
Q2 metal loading	0.00	112.14
Flow reference	5.09	10.93
Silica temperature coordinates	184.68	192.50
Flow-stop relaxation	1.10	4.63

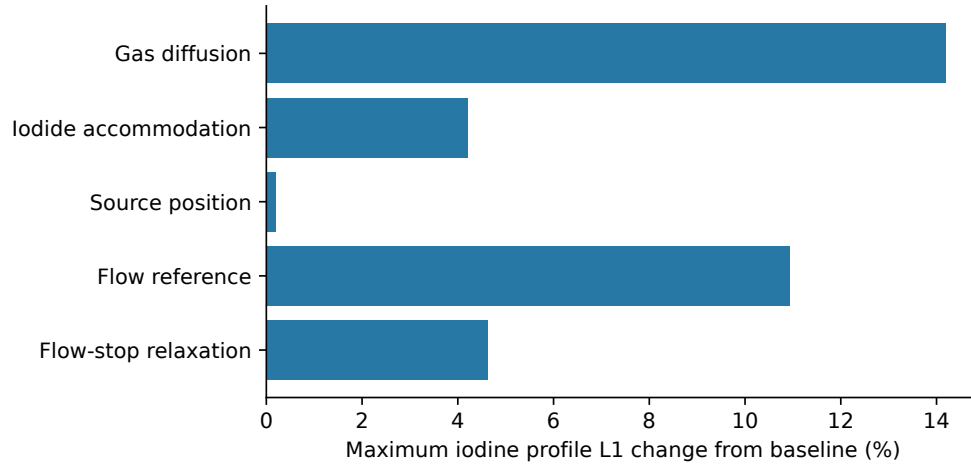


Figure 7: Maximum baseline-relative iodine profile changes among the selected physical-input comparisons. Flow-stop relaxation uses an 80 s endpoint; the others use 60 s. These maxima describe the tested case set, not a general uncertainty budget or an experimentally identified parameter ranking.

The four fixed-temperature flow-stop cases change deposited iodine profiles by 1.10–4.63% over 20 s, with centroid shifts no larger than 0.0422 cm. They show that residual gas can alter wall inventories after release stops. The column is neither cooling nor supplied with a clean-helium flush in these runs. Consequently, they do not predict the actual end-of-experiment redistribution after the furnaces are turned off. Likewise, equilibrium speciation and the irreversible BiI wall channel remain unqualified kinetic approximations.

8 Conclusions

A conservative finite-volume treatment of nine Pb–Bi–I gases and four shared condensed wall inventories has been exercised in 15 baseline and 64 follow-up cases. The reported calculations use a specialised ideal-gas equilibrium kernel, a prescribed helium carrier and reversible iodide wall exchange with half-cell diffusion resistance. Material bookkeeping is strong: the worst raw source-driven follow-up element residue is 8.85×10^{-13} of supplied material. Conservation is a numerical property of this model, rather than evidence that all physical pathways have been represented.

The isolated time-step comparisons give iodine profile changes below 0.323%, while medium-to-fine mesh comparisons give 2.06–10.58%. Stable total deposition and small centroid changes therefore do not establish spatial convergence of local profiles. Closely competing 1 cm bins can also make a reported peak temperature switch by tens of kelvin under refinement. Further spatial and full-run iteration checks are required before claiming precise local predictive accuracy.

Thermodynamic and source assumptions materially affect the computed iodide partition. On the refined steel case, a +29 kJ/mol BiI gas Gibbs-energy surrogate changes the BiI₃ iodine share from 94.57% to 2.00%. This sensitivity survives refinement but does not determine which thermodynamic branch represents the experiment. Q1 agreement

Table 7: Transported gas molar masses. Condensed iodide reservoirs use the same formula masses; shared Pb and Bi inventories use atomic masses.

Gas	kg/mol	Gas	kg/mol	Gas	kg/mol
Pb	0.207200	Bi	0.2089804	I	0.12690447
PbI	0.33410447	Bi ₂	0.4179608	I ₂	0.25380894
PbI ₂	0.46100894	BiI	0.33588487	BiI ₃	0.58969381

is conditioned on inferred deposit masses, and Q2 retains a deposit-derived iodine source and an unmeasured metal saturation.

For silica, input-derived temperature-coordinate shifts reduce the original 4–5 cm position discrepancies to 0–1 cm. The improvement identifies coordinate registration as a consequential uncertainty; it does not establish a corrected experimental dataset. The supplied paper does not resolve that registration or the flow-reference conditions.

The present evidence supports a methods and input-sensitivity study. An independent predictive validation would require better-constrained source and thermodynamic inputs, consistent experimental coordinates, spatially qualified profile calculations and treatment of the three-hour release and cooling protocol. The completed cases identify where that additional evidence is needed without hiding the original profile discrepancies.

Data and software availability

The solver and benchmark preparation scripts are maintained in the LESTO repository, <https://github.com/JanKren/LESTO>. The accompanying working-manuscript files contain the completed baseline and follow-up metrics, saved axial profiles, generated tables, figure-generation code and SHA-256 provenance records. Experimental curves were digitised from Liu et al. [1]; they are reconstructions of published figures, rather than author-supplied measurements. Figure data are adapted under the source paper’s CC BY 4.0 licence. Some thermodynamic coefficient files are external inputs and are not redistributed. Their local identifiers and derived-table provenance are recorded with the benchmark. Reproducing the CFD calculation from coefficients requires access to those inputs; regenerating the manuscript figures uses the saved results and does not require them.

A Properties, thermodynamic provenance and units

A.1 Helium and transported molecular masses

The executed helium carrier uses $M_{\text{He}} = 0.0040026 \text{ kg mol}^{-1}$, $C_p = 5193 \text{ J kg}^{-1} \text{ K}^{-1}$ and

$$\rho = \frac{pM_{\text{He}}}{RT}, \quad \mu(T) = \frac{A_s\sqrt{T}}{1 + T_s/T}, \quad A_s = 1.4683 \times 10^{-6}, \quad T_s = 83.40 \text{ K}. \quad (20)$$

The numerical value of A_s gives viscosity in Pa s when T is in K. These are case inputs to the perfect-gas, constant-heat-capacity, Sutherland model. They are not newly fitted measured helium properties. Pressure used in partial-pressure conversions is absolute, while thermodynamic activities use $p^\circ = 10^5 \text{ Pa}$. The assumed controller reference pressure, 101,325 Pa, is a separate quantity. Table 7 records the species masses used in transport and stoichiometric conversion.

A.2 Thermodynamic records and phase treatment

Pb–I–He functions are evaluated from external NASA-nine-coefficient records labelled with Gurvich source information. Bi–I functions use an external ThermoFun compilation labelled HERACLES/Barin. The common property evaluator converts records to a consistent absolute $G^\circ(T) = H^\circ(T) - TS^\circ(T)$ convention before forming reaction Gibbs energies; mixing this convention with an apparent formation-Gibbs value would give inconsistent equilibrium constants. Tables used by the solver store formation constants and phase-resolved vapour pressures logarithmically. Input-file hashes, record identifiers and table-generation scripts are recorded in the benchmark provenance.

For an iodide with one gas molecule per condensed formula unit,

$$p_{\text{eq}}(T) = p^\circ \exp \left[-\frac{G_g^\circ(T) - G_c^\circ(T)}{RT} \right], \quad (21)$$

using the stable available condensed branch. The BiI_3 liquid branch is constructed at $T_{\text{fus}} = 680.85 \text{ K}$ from the crystalline record with $\Delta H_{\text{fus}} = 31.798 \text{ kJ mol}^{-1}$ and $C_{p,l} = 150.624 \text{ J mol}^{-1} \text{ K}^{-1}$, as recorded in the input-generation metadata. It is a constructed phase record, rather than an independently confirmed liquid thermodynamic dataset. Bi and Bi_2 share condensed Bi but have different gas chemical potentials and stoichiometry.

The $\text{BiI} + 29 \text{ kJ/mol}$ variant alters the gas record and regenerates all formation and disproportionation constants consistently. Baseline and perturbed functions are hypotheses for the sensitivity calculation; independent confirmation of the BiI source record remains needed. External coefficient files are not reproduced in the manuscript data package. Numerical conservation of their derived model cannot establish their physical correctness or complete the provenance assessment.

A.3 Optional PbI_2 –He diffusion estimate

Only the PbI_2 –He transport variant uses the following implemented correlation:

$$D_{\text{PbI}_2\text{--He}} = \frac{1.86 \times 10^{-7} T^{3/2} \sqrt{1/4.0026 + 1/461}}{(p/101325) \sigma^2 \Omega(T^*)}, \quad (22)$$

$$T^* = T/71.117857, \quad \sigma = 4.258468, \quad (23)$$

$$\Omega(T^*) = 0.5313 - 1.2946e^{-0.9922T^*} + \frac{1.5591}{T^*} - \frac{0.1918}{(T^*)^2}. \quad (24)$$

In this numerical expression T is in K, p in Pa, σ in Å and the mass values in g/mol; the resulting D is in $\text{m}^2 \text{ s}^{-1}$. The implementation retains 461 g/mol to reproduce the original correlation, rather than the more precise transport formula mass in table 7. Its estimated collision parameters use the original polarizability/electron-count construction recorded in the code. Other eight gases retain the constant baseline coefficient in this variant. The correlation is a screening alternative, not an experimentally validated binary diffusivity or a replacement for all missing Pb–Bi–I transport data.

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